

FAQs

Q1: Why not rely solely on ARIMA models for trading strategies?

A1: ARIMA captures mean returns but **assumes constant volatility**. Financial **returns show volatility clustering**, which **ARIMA alone cannot model**. Hence, volatility models like ARCH/GARCH are used.

Q2: What is the difference between ARCH and GARCH models?

A2: The ARCH model models volatility based only on the **most recent squared return**. The GARCH model is more general; it also **includes past volatility estimates**.

Q3: Why are transaction costs important?

A3: Backtests often ignore these costs and **look profitable, but turn out to be less profitable** and sometimes even loss-making in practice. Incorporating bid-ask spreads, brokerage, taxes, commissions, etc., gives a more realistic picture of strategy performance.

Q4: Can automated model selection replace manual parameter tuning?

A4: Tools such as *sktime* usually **accelerate parameter search** but risk **overfitting**. **Human oversight** is needed for validating results across assets and regimes.

Q5: How are volatility forecasts used in trading strategies?

A5: Volatility forecasts help in **position sizing, risk management, and options trading**.

Q6: What are integrated ARIMA+GARCH models?

A6: These models combine **ARIMA for direction** and **GARCH for position sizing**. This approach captures **both expected returns and changing risk**, offering a more comprehensive framework for trading.

Q7: How do ARCH/GARCH models improve strategy robustness?

A7: By explicitly modeling volatility clustering, these models provide **realistic risk-adjusted forecasts**, helping strategies perform across **different market conditions**.